

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 31-08-2026

1. TITLE

Design of a Basic Webpage using HTML Tags

2. AIM

To develop a webpage displaying college information utilizing fundamental HTML tags such as headings, paragraphs, links, lists, tables, images, forms, and multimedia elements.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
\n
- To implement text formatting, hyperlinks, and multimedia tags.
\n
- To organize tabular data using HTML tables.
\n
- To build a user input form for student registration.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language for creating web pages. It provides structural semantics for text and media. Elements like <header>, <nav>, <section>, <article>, and <footer> offer semantic meaning, while tags like <table>, , <audio>, and <video> enable embedding of data and media. Forms allow collection of user input via various input types.

5. ALGORITHM / PROCEDURE

1. Create index.html and define the basic document structure.
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2. Add a header and navigation links to different sections.
\n
3. Use text formatting tags (bold, italic, sub, sup) in the about section.
\n

4. Implement ordered and unordered lists for courses and facilities.
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5. Add internal and external hyperlinks using the anchor tag.
 \n
6. Design a student details table with borders and cell padding.
 \n
7. Embed images for the college logo and campus.
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8. Create a registration form with various input types (text, email, date, radio, select).
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9. Add audio and video controls for multimedia playback.
 \n
10. Test and verify the page structure in a browser.

6. PROGRAM

index.html

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<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>

<body>

  <!-- ===== HEADER ===== -->
  <header>
    <h1>ANITS - College Information Portal</h1>

    <p>
      Welcome to <b>Anil Neerukonda Institute of Technology and Sciences (ANITS)
</b>,
      Visakhapatnam.
    </p>
  </header>

  <!-- ===== NAVIGATION ===== -->
  <nav>
    <a href="#about">About</a> |
    <a href="#courses">Courses</a> |
    <a href="#students">Students</a> |
    <a href="#register">Registration</a>
  </nav>

  <hr>

  <!-- ===== SECTION ===== -->
  <section id="about">

    <article>

      <h2>About ANITS</h2>

      <p>
        <b>ANITS</b> is one of the premier engineering colleges in Andhra
Pradesh.
        It offers quality education with excellent laboratories,
        experienced faculty and placement opportunities.
      </p>

      <h3>Text Formatting Example</h3>

      <p>
        <b>Vinay</b> (boldtext) <br>
        <i>Vijay</i> (Italic Text) <br>
        <u>jyothi</u> (Underlined Text) <br>
        H<sub>2</sub>O (Subscript) <br>

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        x<sup>2</sup> + y<sup>2</sup> (Superscript)
    </p>

</article>

</section>

<hr>

<!-- ===== LISTS ===== -->

<section id="courses">

    <h2>Courses Offered (Ordered List)</h2>

    <ol>
        <li>CSE</li>
        <li>CSE (AI&ML)</li>
        <li>CSD</li>
        <li>MECH</li>
        <li>EEE</li>
    </ol>

    <h2>College Facilities (Unordered List)</h2>

    <ul>
        <li>Central Library</li>
        <li>Computer Labs</li>
        <li>Hostel</li>
        <li>play ground</li>
        <li>Transport Facility</li>
    </ul>

</section>

<hr>

<!-- ===== HYPERLINKS ===== -->

<h2>Useful Links</h2>

<a href="https://anits.org/" target="_self">
    Visit ANITS Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
    Visit Google
</a>

<br><br>

<a href="https://www.youtube.com" target="_parent">

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    Visit youtube
</a>
<hr>

<!-- ===== TABLE ===== -->

<section id="students">

    <h2>Student Details</h2>

    <table border="1" cellpadding="10">

        <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Branch</th>
            <th>Email</th>
        </tr>

        <tr>
            <td>101</td>
            <td>Vinay</td>
            <td>CSE (AI&ML)</td>
            <td>vinay@gmail.com</td>
        </tr>

        <tr>
            <td>102</td>
            <td>Santhosh</td>
            <td>CSE</td>
            <td>santhosh@gmail.com</td>
        </tr>

        <tr>
            <td>103</td>
            <td>Vijay</td>
            <td>MECH</td>
            <td>vijay@gmail.com</td>
        </tr>

    </table>

</section>

<hr>

<!-- ===== IMAGES ===== -->

<h2>College Logo</h2>



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<br><br>

<h2>College Campus</h2>



<hr>

<!-- ===== FORM ===== -->

<section id="register">

    <h2>Student Registration Form</h2>

    <form>

        <label>Name:</label><br>
        <input type="text" placeholder="Enter your name">
        <br><br>

        <label>Email:</label><br>
        <input type="email" required placeholder="Enter your email">

        <br><br>

        <label>Password:</label><br>
        <input type="password">

        <br><br>

        <label>Date of Birth:</label><br>
        <input type="date">

        <br><br>

        <label>Age:</label><br>
        <input type="number">

        <br><br>

        <label>Favourite Color:</label><br>
        <input type="color">

        <br><br>

        <label>Programming Skill:</label><br>
        <input type="range">

        <br><br>

        <label>Gender:</label><br>

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<input type="radio" name="gender"> Male

<input type="radio" name="gender"> Female

<input type="radio" name="gender"> Other

<br><br>

<label>Hobbies:</label><br>

<input type="checkbox"> Reading

<input type="checkbox"> Coding

<input type="checkbox"> Sports

<br><br>

<label>Branch:</label>

<select>

    <option>CSE</option>

    <option selected>CSE (AI&ML)</option>

    <option>ECE</option>

    <option>EEE</option>

    <option>IT</option>

</select>

<br><br>

<label>Comments:</label><br>

<textarea rows="5" cols="40"></textarea>

<br><br>

<input type="submit" value="Submit">

<input type="reset" value="Reset">

</form>

</section>

<hr>

<!-- ===== AUDIO ===== -->

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<h2>College Audio</h2>

<audio controls>

    <source src="college.mp3" type="audio/mpeg">

</audio>

<hr>

<!-- ===== VIDEO ===== -->

<h2>College Video</h2>

<video width="500" controls>

    <source src="https://youtu.be/LzKjue8PchA" type="video/mp4">

</video>

<hr>

<!-- ===== FOOTER ===== -->

<footer>

    <p>
        © 2026 ANITS | College Information Portal
    </p>

</footer>

</body>

</html>

```

7. CODE EXPLANATION

Code / Syntax	Explanation
<!DOCTYPE html>	Defines the document type as HTML5.
<a>	Anchor tag used to create hyperlinks.
<table>	Defines a table for tabular data.
<form>	Contains interactive controls for submitting user data.
<video>, <audio>	Embeds media players.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Form Elements	Verify all input fields	Form fields render correctly	Pass
TC-02	Hyperlinks	Click on links	Links open correct targets	Pass
TC-03	Multimedia	Play audio/video	Media plays successfully	Pass

9. OUTPUT

Output 1: Design of a Basic Webpage using HTML Tags

Screenshot not found: observation tasks/task 1\result.png

10. RESULT

Thus, a comprehensive HTML webpage incorporating text formatting, lists, tables, forms, and multimedia elements was successfully designed and verified.

Full Stack Web Development Lab (23CM4121) | K.Vinay - A24126552086